

Oil & Gas Risk and Production Overview

Northeast India and Rajasthan — Simplified Version

Purpose: A simple explanation of the supplied cases, risks, abandoned exploration, and successful production projects.

1. Northeast India (Assam & Tripura)

■ Red — High Risk: Blowouts & Uncontrolled Gas

The Cases: High-pressure gas wells, especially the **Baghjan field** in Tinsukia district, Assam.

The Details: High-pressure gas underground can cause a blowout if pressure is not controlled. The Baghjan #5 well had a serious pressure-control failure, resulting in a large uncontrolled gas leak and fire. The incident continued for months and caused major environmental damage.

■ Yellow — Mid Risk: Monsoons & Aging Infrastructure

The Cases: Mature oil fields in Assam, such as **Geleki**, and remote gas fields in Tripura, such as **Khubal**.

The Details: Heavy monsoon rain can flood roads and make remote fields difficult to reach. Older wells also need regular maintenance. Problems with loose sand and underground water can damage or flood pipes and equipment.

■ Brown — Abandoned: Failed Exploration

The Cases: Exploratory dry holes across the **Assam-Arakan Fold Belt**.

The Details: Explorers drilled deep into complicated rock formations looking for trapped oil. Some wells found mainly water, very little gas, or rock that did not allow hydrocarbons to flow properly. These wells were then permanently plugged and abandoned.

■ Green — Active & Successful: Current Production

The Cases: Historic and active oil fields in Assam, such as **Digboi** and **Naharkatiya**, and major gas-producing areas in Tripura, such as the **Agartala Dome**.

The Details: Assam continues to produce crude oil from active fields. Tripura produces natural gas that is transported by pipeline to local power plants, where it is used to generate electricity.

2. Rajasthan — Barmer (Aishwariya Barmer Hill Project)

■ Red — High Risk: Well Casing Failures

The Cases: Deep, high-pressure extraction wells in the **Aishwariya Barmer Hill (ABH) field**.

The Details: The oil is trapped in very tight rock. Operators use high-pressure hydraulic fracturing to help the oil move through the rock. If steel well casings cannot withstand the pressure, they may fail underground or create dangerous pressure problems at the surface.

■ Yellow — Mid Risk: Desert Water Scarcity & Tight Rock

The Cases: Fracking and horizontal drilling operations across the wider **Barmer District**.

The Details: The rock is very dense, so extraction requires large quantities of water and sand for hydraulic fracturing. Getting and transporting this water in the dry Rajasthan desert creates a major logistical challenge.

3. Rajasthan — Abandoned and Successful Cases

■ Brown — Abandoned: Un-frackable Wells

The Cases: Early appraisal and exploratory wells drilled into the **Barmer Hill formation**.

The Details: Operators knew oil was present, but the rock was extremely dense and did not have enough natural fractures. Even after applying hydraulic-fracturing methods, some wells did not produce enough oil to be profitable. These wells were therefore abandoned as uncommercial.

■ Green — Active & Successful: Advanced Extraction

The Cases: Currently producing horizontal wells in the **Aishwariya Barmer Hill Project**, operated by Cairn Oil & Gas.

The Details: Horizontal drilling turns the well path sideways underground. Combined with multi-stage hydraulic fracturing, this technology helps operators produce oil from very tight rock that was previously difficult to develop commercially.

4. Quick Summary

Region	Risk / Status	Main Point
Assam & Tripura	Red	High-pressure gas wells can face blowout and uncontrolled-gas risks.
Assam & Tripura	Yellow	Monsoon flooding and aging equipment create operational problems.
Assam & Tripura	Brown	Some exploratory wells were dry or non-commercial and were abandoned.
Assam & Tripura	Green	Assam produces crude oil; Tripura produces gas for power generation.
Rajasthan / Barmer	Red	High-pressure extraction can put heavy demands on well casings.
Rajasthan / Barmer	Yellow	Tight rock and water scarcity make drilling and fracking difficult.
Rajasthan / Barmer	Brown	Some early wells were abandoned because production was not commercially viable.
Rajasthan / Barmer	Green	Horizontal drilling and hydraulic fracturing support current production.

5. Frontend / Database Mapping

To integrate these specific location statuses, risk profiles, and project names into your database or UI mapping system, reference the file named "**What actually i have to do for my web**".

Content note: This PDF simplifies and formats the information supplied in the request. The factual claims, project names, and technical details were not independently verified during this conversion.